

AI-Based Migraine Symptom Classifier for Early Detection and Triage

Background:

You are a **Data Scientist at NeuroHealthTech**, a digital healthcare company focused on building AI-powered solutions to support clinical decision-making in neurology. One of the company’s key products is a smart triage assistant to help general physicians assess and refer migraine patients more accurately.

Migraines affect over one billion people worldwide and come in several clinical subtypes, such as migraines with aura, without aura, and other rare types. Delayed diagnosis or misclassification can lead to ineffective treatment and reduced quality of life. NeuroHealthTech aims to solve this problem using data-driven AI models.

To kickstart this initiative, your team acquired a **publicly available dataset** from **Kaggle**, created by Gözde Kızılkaya, which contains structured symptom records for different types of migraines. Your goal is to use this dataset to build and deploy a **Machine Learning-based migraine classification model**.

Data Source:

- **Dataset Name:** [Migraine Symptom Dataset for Classification](#)
- **Author:** Gözde Kızılkaya
- **Format:** CSV (migraine_symptom_classification.csv)
- **Size:** ~2000 records
- **Use License:** Open for research and non-commercial use

Dataset Overview:

Each row in the dataset represents a patient's migraine episode based on self-reported symptoms. The dataset includes:

Dataset Description:

The dataset contains detailed features about migraine episodes reported by individuals. Each record includes:

Column Name	Description
Age	Age of the participant (numeric)
Duration	Duration of the migraine attack (hours)
Frequency	Number of attacks over a given period

Location	Pain location: 1 - Unilateral, 2 - Bilateral, 3 - Frontal, 4 - Temporal
Character	Pain nature: 1 - Throbbing, 2 - Pressing, 3 - Sharp, 4 - Dull
Intensity	Pain severity from 1 (mild) to 5 (very severe)
Nausea, Vomit, Phonophobia, Photophobia, Visual, Sensory, Motor, Language, Vertigo, Tinnitus, Hypoacusis, Diplopia, Defect, Ataxia, Conscience, Paresthesia	Binary indicators (1: Yes, 0: No) of various neurological and systemic symptoms
DPF	Dysfunctional physical factor (e.g., menstruation as trigger)
Type	🏷️ Target variable: Type of migraine diagnosed (e.g., "Typical aura with migraine", "Migraine without aura", etc.)

Target Classes:

- Migraine without aura
- Migraine with aura
- Chronic migraine
- Hemiplegic migraine
- Retinal migraine

Objective:

Develop a supervised machine learning model to **predict the type of migraine** based on symptom features, for use in clinical triage and digital health tools.

Tools & Technologies:

Area	Tools/Frameworks
Language	Python 3.x
IDE	Jupyter Notebook / PyCharm
EDA	Pandas, Seaborn, Matplotlib, Plotly

ML Modeling	Scikit-learn, XGBoost, LightGBM, or Keras (optional)
Model Evaluation	Classification metrics, SHAP
App Deployment	Streamlit (optional for interactive demo)
Model Persistence	joblib or pickle

Project Workflow:

1. Data Exploration (EDA)

- Analyze class distribution of Type
- Visualize relationships between symptoms and migraine types
- Identify correlations and possible data quality issues

2. Data Preprocessing

- Encode categorical variables (Location, Character)
- Normalize or scale numeric features (Age, Duration, Frequency)
- Handle class imbalance (e.g., using SMOTE)
- Feature selection or dimensionality reduction (optional)

3. Model Building

- Baseline classifiers: Logistic Regression, Random Forest
- Advanced models: XGBoost, LightGBM, or a simple neural net (Keras)
- Hyperparameter tuning using GridSearchCV or Optuna

4. Model Evaluation

- Use metrics: Accuracy, F1-score, Confusion Matrix
- Cross-validation for generalization
- Explainability: SHAP or feature importance plots

5. Optional: Streamlit App

- Symptom form for user input
- Model returns predicted migraine type
- Show confidence scores and explanation (e.g., top 3 symptoms contributing to the decision)

6. Model Deployment

- Save the final model using joblib
- Document model assumptions and performance
- Prepare README and Streamlit script for demo

Suggested Use Cases:

- **Multiclass classification** of migraine types from symptom data
 - **Interactive EDA and visualization** for neurology researchers
 - **Feature engineering and modeling practice** with clinical data
 - **Prototype development** for AI triage or telemedicine platforms
 - **Medical education** use for neurology or data science students
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Sample Use Case (Streamlit Interface):

A nurse enters the following patient symptoms into the Streamlit tool:

- Age: 32
- Duration: 6 hours
- Location: 1 (Unilateral)
- Nausea: Yes
- Visual: Yes
- Motor: No

The app predicts:

"Typical aura with migraine" (Confidence: 91%)